

Tnews Widget — Project Documentation

iOS app · Capacitor · WidgetKit · v2.0.1

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Repository: github.com/ziedsG1/Tnewsipa

Current iOS version: 2.0.1

Bundle ID: `tn.tnews.widget`

Display name: Tnews Widget

This document describes the full project: purpose, architecture, features, how to build and install, and how to maintain the codebase.

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1. Overview

Tnews Widget is a Tunisia-focused news app in Arabic (RTL). It was ported from the original **Windows Electron desktop widget** (`C:\Program Files\Tnews Widget`) to **iOS** using [Capacitor](#).

The app:

- Fetches headlines from multiple Tunisian RSS feeds
- Shows a scrollable list of news cards in the app
- Syncs headlines to an **iOS home-screen widget** (WidgetKit)
- Adds mobile-only features: weather, dark/light theme, story-style image sharing, and local notifications

You **cannot build a signed .ipa on Windows alone** — Apple requires macOS and Xcode. This repo includes a **GitHub Actions** workflow that builds an unsigned IPA on a Mac runner.

2. Features

Feature	Description
News list	Scrollable cards with source, topic, time, and summary
RSS sources	Nawaat, Al Katiba (priority), TAP, La Presse, Mosaïque FM, Business News, Webdo.tn
Caching	<code>localStorage</code> cache; stale cache used if network fails
Background refresh	Every 3 minutes while app is open
Home-screen widget	Small / medium / large WidgetKit views; rotates headlines
Weather	Open-Meteo for Tunis (no API key)
Theme	Dark / light toggle, persisted
Share	1080×1920 story image (canvas) for Instagram / Facebook; per-article ↗ on cards
Open article	Double-tap a card to open source URL in in-app browser
Notifications	Optional local notifications (title + relative time) when news updates
Refresh	Header ↻ reloads news and weather

3. Architecture

```
flowchart TB
    subgraph Web["www/ (Capacitor WebView)"]
        HTML[index.html]
        WJS[widget.js - UI]
        AJS[app.js - cache & bridge]
        NF[newsFetcher.js]
        WTH[weather.js]
        SC[shareCard.js]
        NT[notifications.js]
    end

    subgraph Native["ios/App (Swift)"]
        WS[WidgetSyncPlugin]
        WDS[WidgetDataStore - App Group]
        WKE[TnewsWidgetExtension - WidgetKit]
        WI[WidgetInstallPlugin - optional guide]
    end

    RSS[(RSS feeds)]
    OM[(Open-Meteo API)]

    HTML --> WJS
    WJS --> AJS
    AJS --> NF
    WJS --> WTH
    WJS --> SC
    WJS --> NT
    NF --> RSS
    WTH --> OM
    AJS -->|saveNews JSON| WS
    WS --> WDS
    WDS --> WKE
```

Layers:

1. **Presentation** — `widget.js`, `styles.css`, `index.html`
2. **Application logic** — `app.js` (cache, refresh schedule, Capacitor plugins)
3. **Services** — `lib/*.js` (fetch, weather, share, notifications)
4. **Native bridge** — Capacitor plugins + custom `WidgetSync` / `WidgetInstall`
5. **Widget extension** — Separate target reading shared App Group data

Scripts load in order (see `index.html`): libraries first, then `app.js`, then `widget.js`.

4. Project structure

```
Tnewsipa/
├── www/                                # Capacitor web assets (copied to ios/App/App/public on sync)
│   ├── index.html                    # Shell, header, news list container
│   ├── styles.css                    # RTL layout, themes, cards
│   ├── widget.js                     # UI: render list, gestures, theme, init
│   ├── app.js                        # News cache, refresh, window.tnewsWidget API
│   └── lib/
│       ├── newsFetcher.js            # RSS fetch + parse + topic inference
│       ├── weather.js                # Open-Meteo + cache
│       ├── shareCard.js              # Canvas story image + native share
│       └── notifications.js          # Local notifications plugin wrapper
├── ios/App/
│   ├── App/                          # Main Capacitor iOS app
│   │   ├── AppDelegate.swift
│   │   ├── Info.plist
│   │   ├── App.entitlements          # App Group: group.tn.tnews.widget
│   │   ├── WidgetSyncPlugin.swift / .m
│   │   ├── WidgetInstallPlugin.swift / .m
│   │   └── Assets.xcassets/         # App icon, splash
│   ├── TnewsWidgetExtension/         # WidgetKit extension
│   │   ├── TnewsWidgetBundle.swift
│   │   └── TnewsWidgetExtension.entitlements
│   ├── Shared/
│   │   └── WidgetDataStore.swift    # Read/write JSON in App Group
│   ├── App.xcodeproj
│   └── Podfile
├── capacitor.config.json             # appId, webDir, plugins
├── package.json                      # Capacitor dependencies & scripts
├── .github/workflows/
│   └── build-ios-ipa.yml             # macOS CI → unsigned IPA artifact
├── extracted-fresh/                  # Reference copy of original Electron app (optional)
├── GITHUB-SETUP.md                  # Step-by-step GitHub + Sideloadly guide
├── README.md                         # Quick start
└── DOCUMENTATION.md                 # This file
```

5. Web application layer

5.1 `window.tnewsWidget` (`app.js`)

Method	Purpose
<code>loadNews()</code>	Returns cached articles immediately if present; refreshes in background
<code>openLink(url)</code>	Opens HTTPS links via <code>@capacitor/browser</code> on device
<code>onNewsUpdated(cb)</code>	Subscribe to background refresh updates
<code>syncToHomeScreenWidget()</code>	Pushes cache to native <code>WidgetSync</code> plugin
<code>share({ title, text, url })</code>	System share sheet
<code>getConfig()</code>	Rotation / window metadata (legacy desktop parity)
<code>minimize()</code>	Native minimize when available

Cache key: `tnews-news-cache` in `localStorage`.

Config (in `app.js`):

Setting	Value
<code>rotateSeconds</code>	8
<code>refreshMinutes</code>	3
<code>maxAgeDays</code>	7 (articles filtered by publish date)

5.2 `window.TnewsNewsFetcher` (`lib/newsFetcher.js`)

- Fetches all feeds in parallel with **10s timeout** per feed
- Uses **Capacitor HTTP** on native iOS when available (avoids WebView CORS limits)
- Falls back to `fetch()` in browser
- Merges items, deduplicates by link, sorts by date
- Assigns Arabic **topic** labels from title keywords (رياضة, اقتصاد, سياسة, ...)
- Priority feeds (Nawaat, Al Katiba) surface first in merge logic

5.3 `window.TnewsWeather` (`lib/weather.js`)

- API: Open-Meteo forecast for Tunis coordinates
- Caches result in `localStorage`
- Returns `{ city, temp, label }` for the header weather bar

5.4 `window.TnewsShareCard` (`lib/shareCard.js`)

- Renders branded 1080×1920 PNG via `<canvas>`
- Saves/shares through `@capacitor/filesystem` and `@capacitor/share` on device

5.5 `window.TnewsNotifications` (`lib/notifications.js`)

- Wraps `@capacitor/local-notifications`
- Toggle via ; schedules a small queue of article notifications
- Called from `app.js` `writeCache()` when notifications are enabled

5.6 UI behavior (`widget.js`)

- **Single tap** on card: no navigation (desktop-style)
- **Double tap** (within 450 ms): open article URL
- **↗ on card**: share that article as story image
- **Header ↗**: share top headline
- **🔇**: toggle `data-theme` on `<html>`
- **🔄**: reload news + weather

6. News and data pipeline

```
RSS feeds → newsFetcher.js → app.js normalize/filter → localStorage
                        ↓
                    widget.js render
                        ↓
WidgetSyncPlugin → App Group → WidgetKit timeline
```

Feed list (in `newsFetcher.js`):

ID	Label	URL
nawaat	نواة — Nawaat	https://nawaat.org/feed/
alqatiba	الكتيبة — Al Katiba	https://alqatiba.com/feed/
tap-tn-ar	TAP	https://www.tap.info.tn/ar/rss/tunisia
lapresse-tn-ar	La Presse	https://www.lapresse.tn/feed/
mosaique-ar	موزاييك	https://www.mosaiquefm.net/ar/rss/
businessnews	Business News	https://www.businessnews.com.tn/rss
webdo-fr	Webdo.tn	https://www.webdo.tn/feed/

If fewer than 3 articles fall within 7 days, the filter widens to 14 days, then falls back to the latest 50 items.

7. iOS native layer

7.1 Capacitor plugins (npm)

- `@capacitor/app`
- `@capacitor/browser`
- `@capacitor/share`
- `@capacitor/filesystem`
- `@capacitor/local-notifications`
- `@capacitor/splash-screen`
- `@capacitor/status-bar`

`capacitor.config.json` enables **CapacitorHttp** for native network requests.

7.2 Custom plugins

Plugin	Role
WidgetSync	<code>saveNews({ payloadJson })</code> writes news JSON to App Group <code>group.tn.tnews.widget</code>
WidgetInstall	Optional helper to explain adding widget manually (Apple does not allow programmatic widget picker)

7.3 Widget extension

- Target: `TnewsWidgetExtension`

- Reads `WidgetDataStore.articles()` from shared UserDefaults
- Timeline rotates up to 16 headlines every 5 minutes per entry; reload policy ~10 minutes
- Supports **systemSmall**, **systemMedium**, **systemLarge**
- RTL layout via SwiftUI `layoutDirection`

Adding the widget on iPhone: Long-press home screen → + → search **Tnews** → choose size. Open the main app at least once so feeds populate the shared cache.

7.4 Entitlements and permissions

- **App Group:** `group.tn.tnews.widget` (main app + extension)
- **Info.plist:** App Transport Security allows network loads; notification usage description; background mode `fetch` where configured

8. Configuration reference

`capacitor.config.json`

Field	Value
<code>appId</code>	<code>tn.tnews.widget</code>
<code>appName</code>	Tnews Widget
<code>webDir</code>	<code>www</code>
<code>ios.backgroundColor</code>	<code>#080c16</code>

`package.json` **scripts**

```
npm install      # Install Capacitor dependencies
npm run sync     # npx cap sync ios - copy www → ios project
npm run open:ios # Open Xcode workspace (requires macOS)
```

Version numbers

- **Marketing version** (user-facing): `MARKETING_VERSION` in `ios/App/App.xcodeproj/project.pbxproj` (e.g. 2.0.1)
- **package.json** `version` may lag; trust Xcode marketing version for IPA builds

9. Development workflow

On Windows (web + sync only)

```
cd c:\Users\zied\Tnewsipa
npm install
# Edit files under www/
npm run sync
git add .
git commit -m "Describe your change"
git push origin main
```

GitHub Actions rebuilds the IPA automatically on push to `main`.

On macOS (full native build)

```
npm install
npx cap sync ios
cd ios/App && pod install
npx cap open ios
```

In Xcode: set **Signing Team**, select iPhone, Run ►.

After any `www/` change

Always run `npx cap sync ios` before committing so `ios/App/App/public` matches.

10. Building and installing on iPhone

Option A — GitHub Actions (recommended from Windows)

1. Push to github.com/ziedsG1/Tnewsipa
2. **Actions** → **Build iOS IPA** → wait for green check
3. Download artifact **TnewsWidget-ipa** → `TnewsWidget-unsigned.ipa`
4. Sign and install with **Sideloadly** (see [GITHUB-SETUP.md](#))

The workflow runs `xcodebuild archive` with signing disabled, then zips `Payload/App.app` into an unsigned IPA.

Option B — Mac + Xcode

Connect iPhone via USB → Run from Xcode (signs with your Apple ID).

Option C — TestFlight

Requires paid Apple Developer Program and App Store Connect upload.

Install notes

Topic	Detail
Unsigned IPA	Must be signed with Apple ID (Sideloadly / AltStore / Xcode)
Free Apple ID	App expires after ~7 days; re-sideload to refresh
Files app	Opening IPA in Files does not install; use Sideloadly from PC
AltStore	Region-dependent; Sideloadly is often easier on Windows

11. User guide

In-app controls

Control	Action
	Refresh news and weather
	Switch dark / light theme
	Enable or disable local notifications (allow iOS permission when prompted)
↗ (header)	Share current top story as image
↗ (on card)	Share that article as story image
Double-tap card	Open article in browser
Scroll list	Browse all loaded headlines

Home-screen widget

1. Install and open the app once (loads news into shared storage).
2. Add **Tnews** widget from the iOS widget gallery.

3. Widget updates when you open the app or when iOS refreshes the widget timeline.

Status messages (Arabic)

Message	Meaning
جاري التحميل...	Initial load
جاري التحديث...	Refresh in progress
لا توجد أخبار — اضغط	No articles; check network and refresh
فشل التحديث — اضغط	Fetch error

12. Troubleshooting

Problem	Likely cause	Fix
Stuck on “جاري التحميل...”	JavaScript parse error in <code>widget.js</code> / <code>app.js</code>	Fix syntax; rebuild IPA; reinstall
Empty news list	No network or all feeds timed out	Wi-Fi/cellular; tap ⌂; cached news should still show if previously loaded
Widget shows placeholder	App never opened or App Group empty	Open main app, wait for headlines, re-add widget
“Unable to install” / integrity error	Unsigned or expired sideload	Re-sign with Sideloadly and Apple ID
App stops after 7 days	Free developer certificate limit	Re-sideload
GitHub workflow fails on <code>pod install</code>	Transient runner issue	Re-run workflow
Notifications silent	Permission denied or toggle off	Settings → Tnews → Notifications; tap 🔔 in app
Weather unavailable	Open-Meteo unreachable	Non-blocking; news still works

Debug tip: Connect device to Safari **Web Inspector** (macOS) → inspect WebView console for `loadNews failed` / `init failed` errors.

13. Related documents

File	Contents
README.md	Short overview and quick start
GITHUB-SETUP.md	GitHub repo creation, Actions, Sideloadly install
capacitor.config.json	Capacitor app metadata
.github/workflows/build-ios-ipa.yml	CI build definition

Origin desktop app

The Windows **Tnews Widget** Electron app lives at `C:\Program Files\Tnews Widget`. Reference extracts in `extracted-fresh/` and `extracted-app-fresh/` mirror its structure (`main.js`, `preload.js`, `src/widget.js`, `lib/newsFetcher.cjs`).

Changelog (high level)

Version	Notes
1.5.x	Initial iOS port from Electron
1.6+	Scrollable news list
1.7	Home-screen widget + install guide
1.8	Weather, theme, share, app icon
1.9	Story share, double-tap to open, source labels
2.0	Local notifications
2.0.1	Loading fix: <code>weatherEmoji</code> syntax, cache-first <code>loadNews</code> , non-blocking init

Last updated for project version 2.0.1.